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REVIEW

Recent advancement in molecular pathways and receptor targeting using natural products for wound healing activity

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Abstract
Wound healing is a physiological process that has multiple phases of inflammation, proliferation, and remodeling that all need to be conducted in a coordinated manner by different cell types, cytokines, and signalling pathways. Recent developments in herbal medicine have shown the therapeutic potential of natural compounds in the augmentation of wound healing by regulation of molecular pathways and receptor targets. This narrative review addresses the wound healing role of herbal medicine using specific examples of growth factor receptors such as Epidermal Growth Factor Receptor, Transforming Growth Factor-β Receptor, Vascular Endothelial Growth Factor Receptor, and Fibroblast Growth Factor Receptor. Immune receptors including Toll-like Receptors and NOD-like Receptors are also involved in regulation of inflammation and immune response during wound healing. In addition, the review addresses the role of neurotransmitter receptors in pain relief and healing of wounds. Some of the herbal extracts, including *Aloe vera, Calendula officinalis, Curcuma longa, honey,* and *Centella asiatica*, have shown very good wound-healing activity owing to their antioxidant, anti-inflammatory, and antimicrobial properties. The comparative analysis of plant-derived and animal-derived natural drugs also focuses on their curative potency in wound healing. Thus, information regarding molecular mechanisms associated with the use of herbal medicine in wound healing may help in the development of novel therapeutic strategies for chronic and non-healing wounds.

Keywords Medicinal plants · Receptor · Wound · Inflammation · Molecular pathway · Signalling

Introduction

Wound healing is a complex physiological process that involves a series of coordinated events that recover structural and functional tissue integrity. The desire to promote nature-based solutions has led to the screening of various plants and herbs for their wound healing potentials with results showing the enhancement of various aspects of the wound healing process (Chinko and Precious-Abraham 2024). Chronic nonhealing wounds remain a considerable challenge in clinical treatment due to excessive inflammation and impeded reepithelialization and angiogenesis. Therefore, the discovery of novel prohealing agents for chronic skin wounds are urgent and important (Li et al. 2024). Skin wounds can vary in severity from small cuts and scratches to burns, lacerations and punctures. Cutaneous wounds can be either acute or chronic. Burns, mechanical trauma and surgical incisions are prominent causes of acute wounds which generally heal more quickly than chronic wounds. Chronic wounds can result in scarring and may require more than three months to heal. These include pressure ulcers, venous ulcers, arterial ulcers and diabetic ulcers. Chronic wounds can be expensive to treat and represent a serious clinical problem (Heidari et al. 2024). Recent techniques that have been developed for the treatment of acute and chronic wounds include tissue transplantation, cell therapy, wound dressings and the use of an instrument (Mirhaj et al. 2022). Wound healing is a complex biological process involving three overlapping but consecutive stages (e.g. inflammation, proliferation and remodeling), which requires the orderly orchestration of various cell types and cytokines (Luo et al. 2024). Nonsteroidal anti-inflammatory drugs (NSAIDs) are the most widely used anti-inflammatory medicines in the world, which usually block Cox-1 and Cox-2 enzymes to inhibit

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prostaglandins expression, and then reduce inflammation and pain. Here, we identified a wound-stress responsive gene, activating transcription factor 3 (ATF3), and then investigated its biological action and mechanism in wound healing. In the full-thickness skin wound model, ATF3 was found to promote fibroblast activation and collagen production, resulted in accelerated wound healing. Mechanically, ATF3 transcriptionally activated TGF-β receptor II via directly binding to its specific promoter motif, followed by the qenhanced TGF-β/Smad pathway in fibroblasts (Luo et al. 2024). TLR/NLR structure at an atomic level would enable in developing antagonist/agonist ligands to modulate fish immunity and disease, thus directly influencing the global economy. Fish, in particular zebrafish, is emerging as a valuable animal model for preclinical in vivo study and drug screening. This further highlights the importance of studying fish innate immune receptors to discover novel therapeutic strategies that can later be tested in higher organisms including humans. Innate immunity driven by pattern recognition receptor (PRR) protects the host from invading pathogens. Aquatic animals like fish where the adaptive immunity is poorly developed majorly rely on their innate immunity modulated by PRRs like toll-like receptors (TLR) and NOD-like receptors (NLR) (Sahoo 2020). The body's natural response to tissue damage is wound healing. However, the vascular system, cytokines, mediators, and a variety of cell types interact intricately during wound healing, making it a complex process. The initial cascade of platelet aggregation and blood vessel vasoconstriction is intended to halt bleeding. A flood of different inflammatory cells, beginning with the neutrophil, follows. A range of mediators and cytokines are then released by these inflammatory cells to encourage thrombosis, angiogenesis, and reepithelialization. In turn, the fibroblasts deposit extracellular materials that will act as scaffolding (Wallace et al. 2023). The inflammatory phase is characterized by hemostasis, chemotaxis, and increased vascular permeability, limiting further damage, closing the wound, removing cellular debris and bacteria, and fostering cellular migration. The duration of the inflammatory stage usually lasts several days (Coger et al. 2019).

During the stages of wound healing, reactive oxygen species are essential for cell signalling pathways. Emerging wound treatment techniques aim to balance the redox level in the deep wound tissue. Natural antioxidants, reactive oxygen species (ROS) scavenging nanozymes, and antioxidant delivery systems are examples of reactive oxygen species scavenging agents that have been widely used in recent years to prevent oxidative stress and encourage skin regeneration (Jain et al. 2021a). This article critically examines the role that reactive oxygen species play in the various stages of wound healing (Fig. 1). Granulation tissue development, reepithelialization, and neovascularisation are characteristics of the proliferative phase. This stage may continue for a few weeks. The wound reaches its maximal strength during the maturation and remodelling period (Bowden et al. 2016).

Receptor mechanisms in wound healing

The process of healing a wound is multifaceted and intricate, including many kinds of cells, growth factors, and signalling networks. Growth factor receptors are essential for coordinating the cellular processes required for efficient wound healing. Four main steps of the process are hemostasis, inflammation, proliferation, and remodeling. Here, we concentrate on the receptor mechanisms necessary for each stage of the healing of wounds (Velnar et al. 2009). During the process of extravasation of inflammatory cells into a wound, important interactions occur between adhesion molecules (selectins, cell adhesion molecules (CAMS) and cadherins) and receptors (integrins) that are associated with the plasma membranes of circulating leukocytes and vascular endothelial cells (Schultz et al. 2011).

Role of growth factor receptors

Through facilitating cellular responses to growth factors—which are necessary for cell proliferation, migration, differentiation, and survival—growth factor receptors play a critical role in wound healing. These receptors play a role in wound healing at different stages, especially in the periods of remodeling and proliferation. This is a thorough examination of the main growth factor receptors and their functions (Demidova-Rice et al. 2012).

a. *Epidermal growth factor receptor (EGFR)*: A transmembrane receptor with tyrosine kinase activity, the epidermal growth factor receptor (EGFR) regulates cellular proliferation, survival, differentiation, and metastasis. In the past 20 years, the development of precise and focused therapy approaches for human cancers has been fuelled by our growing understanding of the intricate function and control of these receptors (Wee and Wang 2017). The first receptor target that monoclonal antibodies (mAb) have been developed against in the therapy of cancer is EGFR. The biology of ErbB receptors is reviewed here, along with its architecture, signalling, regulation, and therapeutic approaches. We also discuss the processes by which the receptors provide resistance to small-molecule tyrosine kinases and mitigate the effects of mAbs (Martinelli et al. 2009). The site-specific extracellular portion of EGFR, which is essential to the dimerization and activation process, determines the effectiveness of EGFR-specific monoclonal antibodies

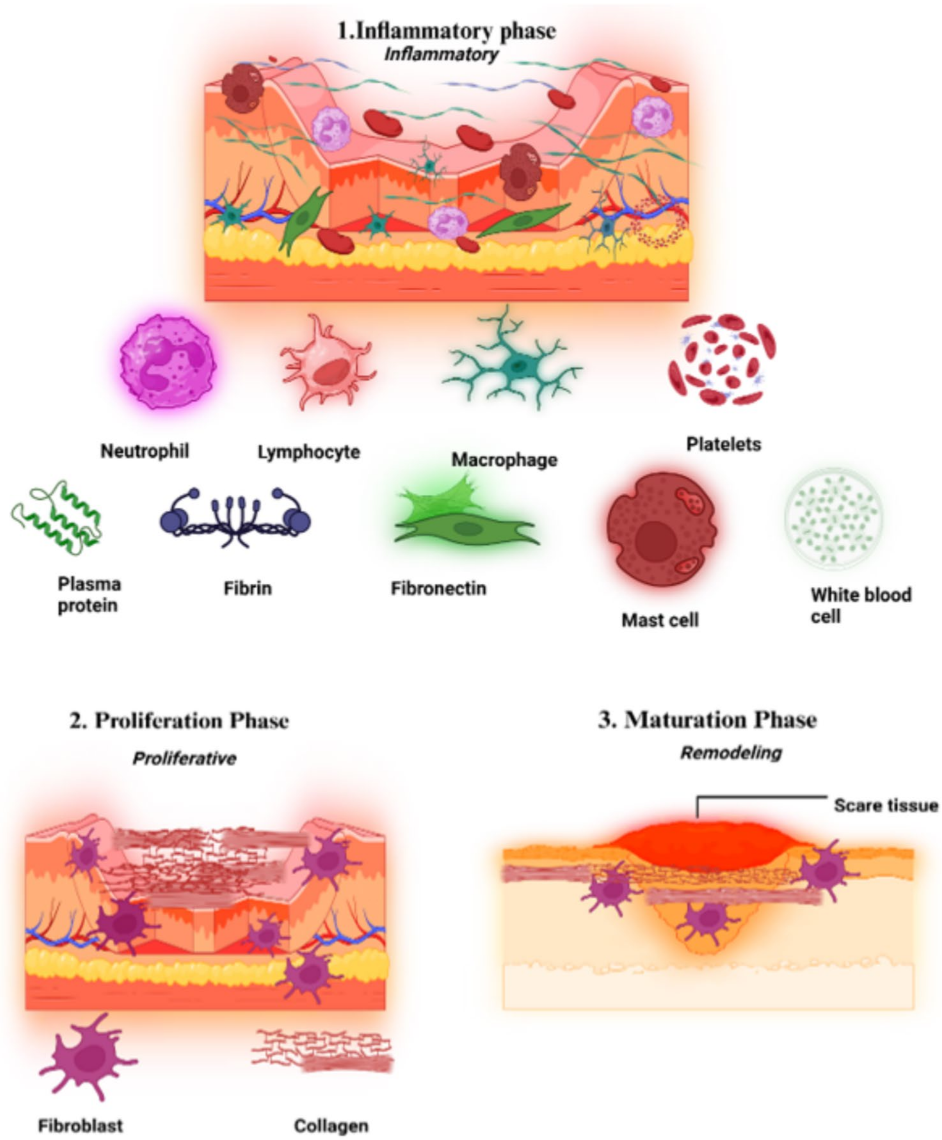


Fig. 1 Skin wound healing is divided into four precisely regulated processes: i.e., hemostasis, inflammation, proliferation, and tissue-remodelling (Li et al. 2023)

in cancer treatment (Fig. 2). This study focuses on how different resistance mechanisms have evolved as a result of the side effects of existing small-molecule anti-EGFR therapy (Dokala and Thakur 2017). The epidermal growth factor receptor, or EGFR, is a key player in the signalling cascade that governs the behavior of epithelial cells and malignancies derived from them. EGFR is a structurally and functionally comparable transmembrane growth factor receptor protein, one of four. Collectively, these receptor proteins includes the family of receptor tyrosine kinases known as c-erbB (Herbst 2004).

b. *TGF-β receptor*: the growth factor TGF-β in the process of both wound healing and scar formation. The three isoforms (TGF-β1, TGF-β2 and TGF-β3) appear to have overlapping functions and predominantly mediate their effects through the intracellular SMAD pathway. Initial research suggested that TGF-β1 was responsible for the fibrotic scarring response whereas the scarless wound healing seen in fetal wounds was due to increased levels of TGF-β3 (Ferguson and MW, O'Kane S. 1445). However, the reality appears to be far more complex and it is unlikely that simply altering the ratio of TGF-β isoforms will lead to scarless wound healing (Penn et al. 2012). Transforming growth factor-β1 (TGF-β1) plays a central role in normal and aberrant wound healing, but the precise mechanism in the local environment remains elusive. Here, using a mouse model of aberrant wound healing resulting in heterotopic ossification (HO) after traumatic injury, we find autocrine TGF-β1 signalling in macrophages, and not mesenchymal stem/progenitor cells, is critical in HO formation. In-depth single-cell transcriptomic and epigenomic analyses in combination with immunostaining of cells from the injury site demonstrated increased TGF-β1 signalling in early

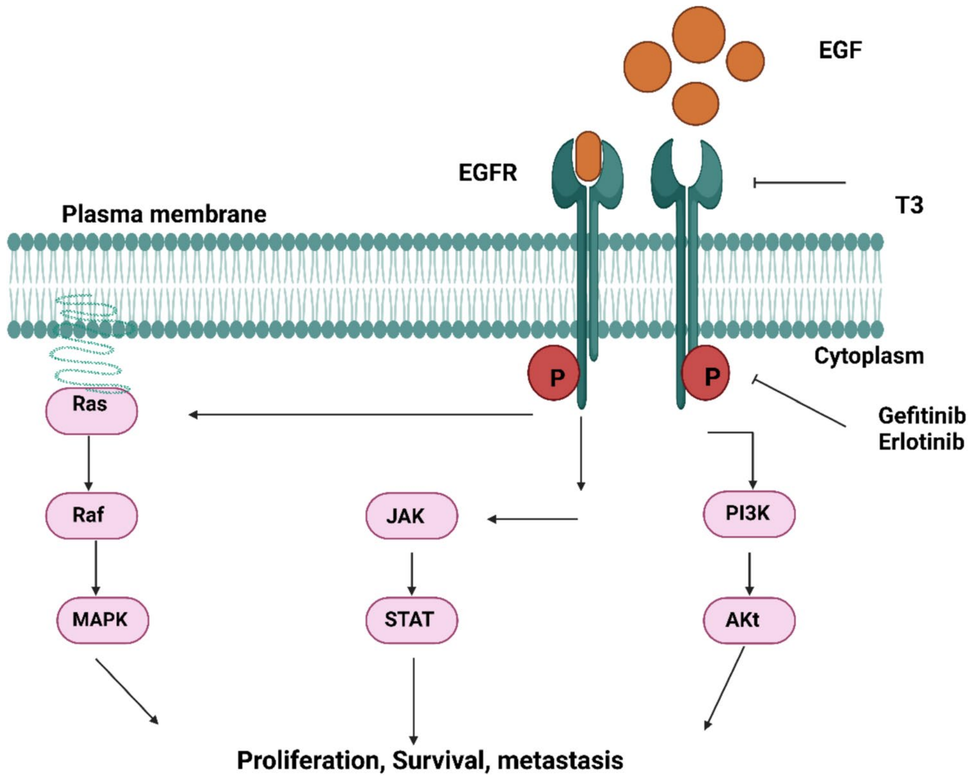


Fig. 2 Schematic representation of EGFR and the signalling proteins. JAK stands for Janus kinase; STAT for signal transducer and activator of transcription; PI3K for phosphatidylinositol 3-kinase; MAPK for mitogen-activated protein kinase; Akt for protein kinase B; P for phosphorylation; and EGF for epidermal growth factor and EGFR for epidermal growth factor receptor. Activation/induction is indicated by arrows, whereas inhibition/suppression is indicated by perpendicular lines (Ferguson and MW, O'Kane S. 1445)

infiltrating macrophages, with open chromatin regions in TGF-β1-stimulated genes at binding sites specific for transcription factors of activated TGF-β1 (SMAD2/3). Genetic deletion of TGF-β1 receptor type 1 (Tgfbr1; Alk5), in macrophages, resulted in increased HO, with a trend toward decreased tendinous HO. To bypass the effect seen by altering the receptor, we administered a systemic treatment with TGF-β1/3 ligand trap TGF-βRII-Fc, which resulted in decreased HO formation and a delay in macrophage infiltration to the injury site. Overall, our data support the role of the TGF-β1/ALK5 signalling pathway in HO (Patel et al. 2022).

By the secretion of cytokines from macrophages, resulted in the proliferation and migration of skin cells and dynamic regulation of TGF-β1 and TGF-β3 in wounds, which accelerated re-epithelialization and granulation tissue formation and thus skin regeneration which shows significant therapeutic potency against chronic wounds, skin fibrosis, and oral ulcers.

c. *Vascular endothelial growth factor (VEGF)* in humans, VEGF binds with receptors Flt-1 (VEGFR-1) and KDR (VEGFR-2), both high affinity receptors. They are members of the Type III tyrosine kinase family, consisting of seven immunoglobulin-like extracellular domains, a single transmembrane spanning domain, and an intracellular tyrosine kinase domain. Two additional receptors have been designated low affinity/molecular mass VEGF receptors, but their structure and function are not well characterized, VEGF stimulates wound healing via multiple mechanisms including collagen deposition, angiogenesis and epithelization. In the clinical setting, the mitogenic, chemotactic, and permeability effects of VEGF may potentially aid to promote repair in nonhealing wounds in arterial occlusive disease and diabetes (Bao et al. 2009). VEGFR-2-activation with VEGF-E increased expression of anti-inflammatory cytokine interleukin (IL)-10 and reduced macrophage infiltration and myofibroblast differentiation in wounded skin compared with controls. VEGF-E treatment also increased microvascular density and improved pericyte coverage of blood vessels in the healing wounds. The ST that formed following treatment with VEGF-E was reduced in size and showed improved collagen structure (Wise et al. 2018).

d. *Fibroblast growth factor receptor (FGFR)*: Fundamentally regulating embryogenesis, angiogenesis, tissue homeostasis, and wound healing is the receptor tyrosine kinase (RTK) signalling pathway known as fibroblast growth factor/fibroblast growth factor receptor (FGF/FGFR) (Chae et al. 2017). Fibroblast growth, angiogenesis, and extracellular matrix deposition are all mediated by FGFRs (Fig. 3). They are essential for tissue healing and wound contraction (Farooq et al. 2021). FGFRs dimerise and autophosphorylate after binding FGFs, triggering the MAPK, PI3K/AKT, and PLCγ pathways that promote cell division, survival, and proliferation

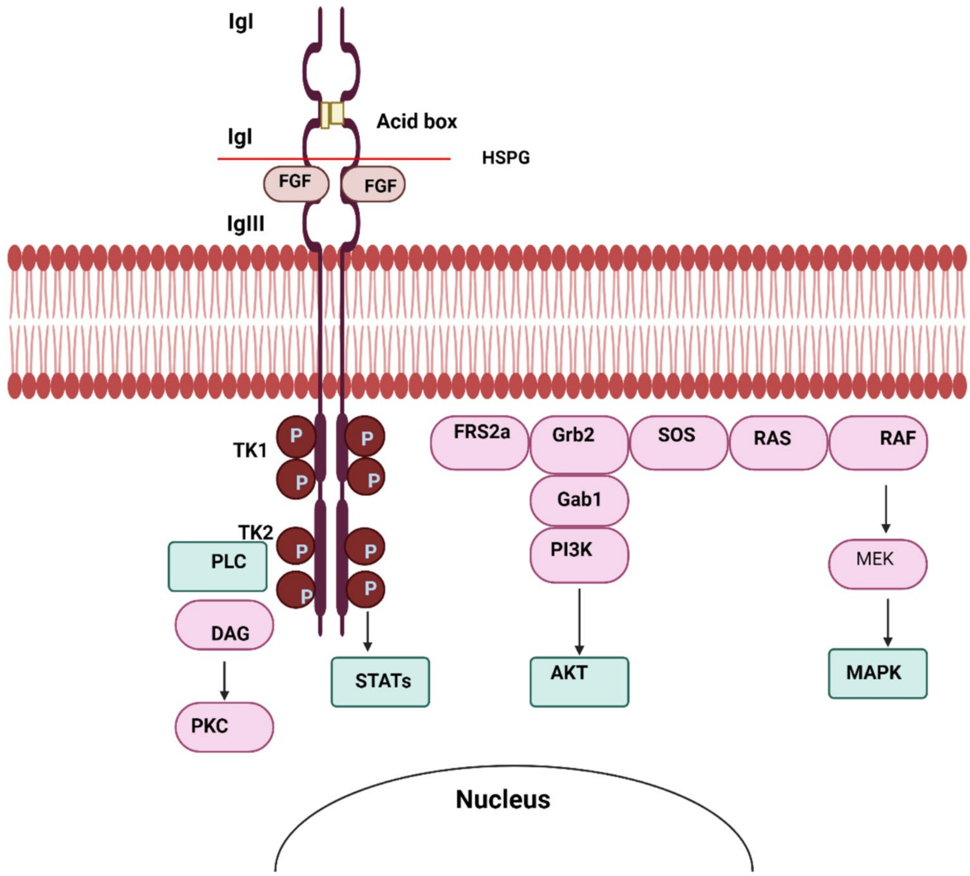


Fig. 3 The breakdown of extracellular matrix (ECM), fibroblast differentiation, collagen remodelling, and macrophage involvement are highlighted in this picture, which depicts the molecular pathways of wound healing. It illustrates how progenitor cells, reticular, papillary, and myofibroblasts interact with mechanical stresses to affect tissue regeneration and extracellular matrix production in the dermal and hypodermal layers (Ellis et al. 2018)

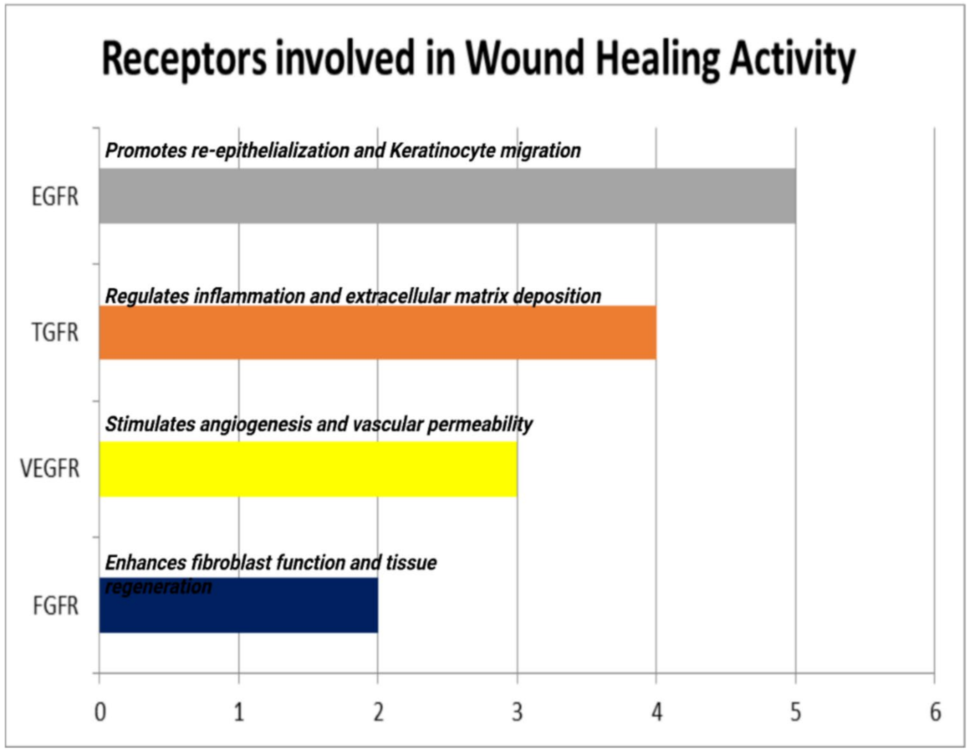
(Ferguson et al. 2021). Fibroblast Growth Factors (FGFs), such as FGF-1 (acidic FGF) and FGF-2 (basic FGF), are ligands (Makrilia et al. 2009).

Molecular abnormalities that cause the FGFR pathway to be activated. When ligands bind to the FGFRs, they dimerise and set off a series of signaling cascades downstream. Gene amplification, translocation, or mutation can all activate the FGFR receptors (Helsten et al. 2015). Activation may also result from an increase in FGF ligands in circulation. Downstream signalling can activate multiple pathways leading to DNA transcription, including the phosphorylation of the signal transducer and activator of transcription (STAT), the phosphoinositide-3-kinase (PI3K/Akt) pathway, the mitogen activated protein kinase (MAPK) pathway, and the PLCγ activation of the DAG-PKC and IP3-Ca2 + cascade. The signaling cascade can be attenuated to variable degrees via negative feedback loops (Fig. 4). As demonstrated above, FGFRs' cytoplasmic domain can interact with members of the "similar expression to FGF" (SEF) family to block downstream signaling. FGFRL1 (atypical receptor/FGFR5) is thought to have three possible roles: it may be a ligand trap; it might dimerise with other transmembrane FGFRs to prevent autophosphorylation; or it could speed up the turnover of other FGFRs (Trueb 2011).

Angiogenesis depends on VEGFR (Vascular Endothelial Growth Factor Receptor), which promotes the development of new blood vessels to supply vital nutrients and oxygen to the regenerated tissue, guaranteeing appropriate metabolic support throughout the healing process (Johnson and Wilgus 2014). In a similar manner, fibroblast growth factor receptor (FGFR) promotes granulation tissue development and fibroblast proliferation, which further supports structural integrity and other angiogenic processes and aids in tissue repair (Yun et al. 2010). Although each receptor serves a distinct purpose, EGFR is notable for its quick response in re-epithelialization, TGFR for controlling inflammatory and fibrotic responses, VEGFR for its specific function in vascular growth, and FGFR for its all-encompassing assistance in connective tissue repair. These receptors work together to ensure a wound healing process that is both effective and balanced. They cooperate to aid in healing (Rayego-Mateos et al. 2018).

The receptor for the vascular endothelial growth factor (VEGFR) by attaching to tyrosine kinase receptors (the VEGFRs) on the cell surface, all members of the VEGF family trigger cellular responses by inducing the receptors to dimerise and transphosphorylate, activating them. The

Fig. 4 FGFR (fibroblast growth factor receptor) structure, signaling, and ligand binding. The structure of the FGF receptor tyrosine kinase is shown schematically. It is made up of multiple domains, such as the acid-box, a transmembrane domain, two intracellular tyrosine kinase domains (TK1 and TK2), and three extracellular immunoglobulin-like domains (IgI, IgII, and IgIII). Two receptor molecules, two FGFs, and one heparan sulphate proteoglycan (HSPG) chain make up the fundamental structure of the FGF/FGFR complex. Four important downstream pathways are triggered by the signal transduction network: PLCγ, PI3K/AKT, STATs, and RAS/RAF/MAPK (Helsten et al. 2015)



VEGF receptors are composed of an intracellular portion that has a split tyrosine-kinase domain, an extracellular portion that contains a single transmembrane spanning region and seven immunoglobulin-likedomains (Shibuya 2011).

Both VEGFR-1 (Flt-1) and VEGFR-2 (KDR/Flk-1) are bound by VEGF-A. It seems that practically all of the identified cellular responses to VEGF are mediated by VEGFR-2 (Rahimi 2006). Though its role is less clear, VEGFR-1 is believed to influence VEGFR-2 signalling. VEGFR-1 also serves as a dummy or decoy receptor, preventing VEGF from binding to VEGFR-2; this role seems to be especially crucial during embryonic vasculogenesis. Actually, sFlt1, an alternatively spliced form of VEGFR-1, is released and mostly serves as a decoy rather than being a membrane-bound protein. There is a third receptor known as VEGFR-3, although VEGF-A is not a ligand for this receptor. In response to VEGF-C and VEGF-D, lymphangiogenesis is mediated by VEGFR-3 (Ceci et al. 2020).

Molecular pathways skin wound healing

Wound healing involves haemostasis, inflammation, proliferation, and remodelling. Platelets clump to stop bleeding, followed by inflammation, fibroblasts, and myofibroblasts. Myofibroblasts tighten wound margins and promote extracellular matrix deposition (Ellis et al. 2018). ECM remodelling is influenced by skin responses, and PRRX1 + fibroblasts release matrix. VEGF stimulates angiogenesis, replenishing blood supply. Myofibroblasts die during remodelling, and MMPs break down extracellular matrix, allowing collagen isoform switching and scar fortification. Mechanical stimuli control fibroblast behavior, but dysregulation can cause persistent wounds. Understanding these pathways aids in developing treatments and regenerative medicine techniques (Bonnans et al. 2014). Fibroblasts, immune cells, and extracellular matrix (ECM) components interact dynamically during the wound healing process to restore tissue integrity. A scab first develops over the wound site, and ECM breaks down to remove injured tissue and promote cell migration. varied fibroblast populations in the dermis have varied functions in tissue repair. In order to restore tensile strength, collagen crosslinking and isoform switching are facilitated by papillary fibroblasts, which are found in the upper dermis. ECM components secreted by reticular fibroblasts, which are located in the deeper dermis, give the regenerating tissue structural support (Tracy et al. 2016). Myofibroblasts are essential for ECM secretion and wound contraction, which promote tissue closure (Fig. 5). They develop from BLIMP-1 + DLK1-progenitors. Macrophages interact with fibroblasts to promote regeneration by modifying their function and removing waste. Mechanical pressures influence fibroblast activity, with EN1 + fibroblasts secreting extracellular matrix in dorsal skin and EN1-fibroblasts repairing ventral skin. PRRX1 + fibroblasts contribute to wound healing. The hypodermis, with BLIMP-1 + DLK1 + progenitors, aids in regeneration with adipocytes, blood arteries,

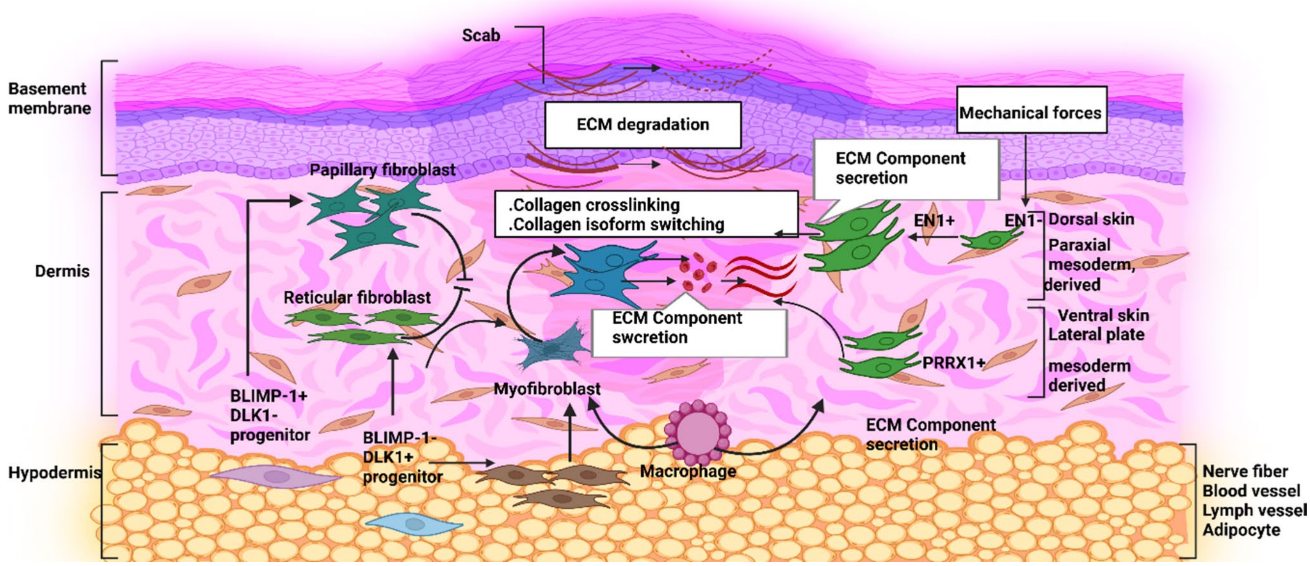


Fig. 5 Numerous receptors, including EGFR, TGFR, VEGFR, and FGFR, are essential for wound healing because they coordinate distinct stages of tissue repair. In order to swiftly re-establish the protective skin barrier, EGFR (Epidermal Growth Factor Receptor) is well known for its speedy action in starting re-epithelialization. It does this by mainly promoting keratinocyte migration and proliferation. By regulating inflammation, encouraging fibroblast activation, and augmenting extracellular matrix deposition for efficient remodelling and scar formation, TGFR (Transforming Growth Factor Beta Receptor) mostly acts in later stages (Farooq et al. 2021; Tito et al. 2025)

lymphatic vessels, and nerve fibers. This process demonstrates the complexity of mechanical signalling, ECM remodelling, and fibroblast heterogeneity (Rodriguez et al. 2019).

Immune receptors and inflammation control

To identify and react to possible threats, the immune system uses an intricate network of receptors. These immunological receptors play a critical role in beginning and controlling the inflammatory response by recognising infections, injured cells, and other toxic chemicals. The main immune receptors and how they affect the regulation of inflammation are summarized TLRs, or Toll-Like Receptors (Chen et al. 2017). Negative feedback mechanisms that guarantee the inflammatory response is proportionate and appropriately resolved, such as the generation of anti-inflammatory cytokines like IL-10 and inhibitors like SOCS (Suppressor of Cytokine Signaling) proteins, delicately regulate TLR signaling. Later, it was shown that Toll plays a crucial role in the generation of an immune defence against bacterial and fungal infections, and similar receptors in vertebrates were found (Yoshimura et al. 2018).

a. *Toll-like receptors (TLRs)*: Toll-like receptors (TLRs) play a pivotal role in wound healing by acting as key sensors of the innate immune system. These proteins are crucial for recognizing pathogen-associated molecular patterns (PAMPs) from bacteria, viruses, and fungi, initiating immune responses that are vital for preventing infections in wounds (Goulopoulou et al. 2016). Upon activation, TLRs trigger the release of pro-inflammatory cytokines, such as TNF-α and IL-6, which recruit immune cells like macrophages and neutrophils to the wound site. This inflammatory phase is essential for clearing debris and pathogens, setting the stage for tissue repair. TLRs also influence the transition from inflammation to the proliferative phase of healing by promoting the activation and differentiation of fibroblasts and keratinocytes. These cells are integral to the formation of new tissue, collagen synthesis, and re-epithelialization. Additionally, TLR activation contributes to angiogenesis, the formation of new blood vessels, by stimulating the release of growth factors such as vascular endothelial growth factor (VEGF), which enhances blood flow and nutrient delivery to the healing tissue (Chen et al. 2017). The therapeutic potential of TLRs in wound healing is increasingly being explored; TLR agonists can be applied topically to boost local immune responses and accelerate healing in chronic wounds. Furthermore, incorporating TLR ligands into biomaterials for wound dressings may enhance their effectiveness by stimulating an appropriate immune response. Overall, TLRs represent a promising target for innovative wound healing strategies, providing insights into how modulating immune responses can facilitate faster and more effective tissue repair. Understanding their complex roles can lead to the development of new therapies that improve outcomes for patients with chronic or non-healing wounds (Sousa et al. 2022; Yamamoto et al. 2011).

b. *NOD-Like receptors (NLRs)*: By modulating inflammation and tissue repair processes, NOD-like receptors (NLRs) are important for wound healing. Damage-associated molecular patterns (DAMPs), which are generated from damaged cells upon tissue injury, are recognised by NLRs, which sets off an inflammatory response that is necessary for wound healing (Roh and Sohn 2018). Preventing infection and laying the groundwork for the ensuing phases of repair and regeneration depend heavily on this inflammatory phase. Moreover, NLRs support the regulation of the equilibrium between tissue regeneration and inflammation, making sure that the latter subsides in a way that permits tissue remodelling and healing (Eming et al. 2014). NLR activity dysregulation can result in persistent inflammation or compromised healing, underscoring their vital function in the process of wound healing (Singh et al. 2023). The nucleotide-binding oligomerization domain-like receptors, or NOD-like receptors (NLRs) (also known as nucleotide-binding leucine-rich repeat receptors), are intracellular sensors of pathogen-associated molecular patterns (PAMPs) that enter the cell via phagocytosis or pores, and damage-associated molecular patterns (DAMPs) that are associated with cell stress. They are types of pattern recognition receptors (PRRs), and play key roles in the regulation of innate immune response. NLRs can cooperate with toll-like receptors (TLRs) and regulate inflammatory and apoptotic response (Yang and Chang 2019).

Neurotransmitter receptors in pain and healing

Neurotransmitter receptors play a critical role in the experience of pain as well as the healing process (Yang and Chang 2019). GABA receptors provide inhibitory control to modulate pain, while glutamate receptors (NMDA and AMPA) and Substance P receptors (NK1) are essential for excitatory pain transmission. Both pain and mood are influenced by adrenergic receptors (beta and alpha) and serotonin receptors (5-HT), with the effects of serotonin differing depending on the receptor subtype. Both dopamine receptors (D1 and D2) and alpha-2 adrenergic receptors are influenced by norepinephrine, which inhibits pain and modifies inflammation (Petroianu et al. 2023). The endocannabinoid system promotes healing by lowering pain and inflammation through CB1 and CB2 receptors. Additionally involved in pain management and immune response modulation are acetylcholine receptors (nicotinic and muscarinic) and opioid receptors (mu, delta, and kappa) (Mathias et al. 2024). Finally, the inflammatory healing response and pain are both mediated by histamine receptors (H1-H4). Gaining an understanding of these receptors aids in the development of focused treatments for improved healing and efficient pain control (Thangam et al. 2018).

Herbal medicine in wound healing

Herbal therapy uses the inherent qualities of many plants to encourage tissue regeneration and lower inflammation, which has a substantial impact on wound healing. Aloe vera promotes cell growth and keeps the healing environment moist thanks to its calming and antibacterial qualities (Hekmatpou et al. 2019). Calendula improves blood flow and collagen synthesis (Ejiohuo et al. 2024), and curcumin (found in turmeric) lowers inflammation and guards against infection (Islam et al. 2024). Due to its antibacterial and moisture-retentive properties, honey helps to minimise scarring and promotes quick healing (Tashkandi 2021). Gotu kola increases the production of collagen and angiogenesis, while the anti-inflammatory properties of chamomile help to relax and rebuild tissue. These herbs are useful for efficient and all-natural wound care since echinacea promotes wound contraction and immunological activity (Gohil et al. 2010).

Aloe vera

Aloe vera a member of the Liliaceae family, is a well-known traditional medicinal plant that has been used for decades to cure a wide range of ailments, from wounds to cancer (Surjushe et al. 2008). It has been incorporated into national pharmacopoeia as well as the traditional and herbal healthcare systems of numerous civilizations worldwide. Its possible anti-inflammatory, wound-healing, and antioxidant properties have also been supported by a number of in vitro and in vivo investigations, all of which are consistent with its traditional and historical applications (Teshome et al. 2022; Anywar et al. 2022). The majority of research, meanwhile, has focused on the latex and gel of A. vera, especially how well it heals wounds. Its bloom has been the subject of very few investigations, and even fewer have examined its molecular mechanism and effects on cutaneous wound healing (Hashemi et al. 2015; Sánchez et al. 2020).

Polysaccharides, vitamin C, vitamin E, lecithin, glycoprotein, anthraquinone, amino acids, and several substances that aid in wound healing are all present in this. Aloe Vera's anti-inflammatory and anti-oxidant qualities are linked to its medicinal qualities (Kaur and Bains 2024). Aloe vera is said to promote collagen formation, activate macrophages, decrease vasoconstriction, oxygenate the wound, inhibit platelet aggregation, and scavenge free radicals, among other possible processes (Ghayempour et al. 2016). Numerous investigations on the use of this plant as a wound-healing substance have been published. Yadav et al. assessed the ability of 50 and 96.4% gel of aloe vera to heal wounds in the experimental rats. According to their findings, Aloe Vera gel has the ability to enhance wound contraction and hasten epithelialisation by influencing the collagen content of the injured tissue (Chelu et al. 2023). Pereira et al. looked at the in vitro degradation rate of calcium alginate hydrogel films containing aloe vera for applications such as drug delivery and wound healing. They demonstrated how the produced hydrogel films' rate of deterioration is greatly accelerated by adding more aloe vera. Curto et al. assessed the impact of aloe vera in vitro on corneal cells' ability to heal wounds. According to their findings, aloe vera can help superficial corneal lesions heal more quickly by accelerating epithelialisation, enhancing collagenase activity, and reducing fibrosis (Pereira et al. 2013). C-glycosyl chromone, a new anti-inflammatory compound, was isolated from the gel extract and may contribute to Aloe vera's anti-inflammatory and anaesthetic effects by inhibiting the cyclooxygenase pathway by reducing prostaglandin E2. Additionally, hydrolysing enzymes from Alo were extracted, including carboxypeptidase and bradykinase. By dissolving bradykinin, which causes pain, these enzymes have strong anti-inflammatory properties (Cosmetic Ingredient Review Expert Panel 2007). The majority of current research on aloe vera's anti-inflammatory properties focusses on the isolated components' mode of action in mice and RAW264.7 mouse macrophage cells that have been activated with lipopolysaccharide (LPS). Therefore, aloin's potential to reduce inflammation is associated with its capacity to decrease the generation of ROS, cytokines, and the JAK1-STAT1/3 signaling pathway (Wang et al. 2023).

Calendula

Folk medicine has utilised the calendula officinalis L., often known as marigold, to treat burns, cuts, and other dermatological conditions. It possesses strong wound-healing, anti-inflammatory, antioxidant, anti-oedematous, antibacterial, antifungal, and antiviral properties (Leach 2008). Terpenoids, polyphenols, coumarins, and other phytochemicals found in calendula flowers are linked to these benefits. Calendula officinalis used topically promotes re-epithelialization and physiological regeneration (Bhattacharjee et al. 2021). European pharmacopoeia officially recognises calendula plants. There are numerous commercially available treatments that contain calendula flower extract, including ointments, lotions, and gels. Adding calendula flower extract (CFE) to natural biomaterials like collagen may improve its ability to heal wounds. Our goal was to create a collagen film that is scientifically transferable, safe, effective, and reasonably priced. This film will support tissue growth while delivering the phytochemicals found in Calendula officinalis flowers to enhance the healing process of wounds (Shahane et al. 2023). Calendula officinalis increased the amount of hexosamine and collagen hydroxyproline content, which significantly improved wound healing activity (Giostri et al. 2022).

Turmeric (Curcuma longa)

One of the ancient herbs that Indonesians have utilised for ages is turmeric, which is thought to provide therapeutic benefits. Because comprehensive turmeric contains proven compounds with antifungal, anti-inflammatory, hepatoprotective, anticancer, and antiviral effects, it is emerging as a novel source of medication to fight a variety of ailments. Turmeric also contains a variety of other components, including tannins, terpenoids, essential oils, alkaloids, and flavonoids in addition to curcumin. It has been demonstrated that curcumin and essential oils have anti-inflammatory qualities (Jain et al. 2024a). The way turmeric extract works on wounds is by blocking the inflammatory enzymes lipooxygenase (LOX) and cyclooxygenase (COX-2), which speed up collagen creation, tissue re-epithelialization, and cell proliferation. Thirteen Turmeric has so far shown to be safe; even at high doses, it doesn't have any noticeable adverse effects (Jain et al. 2024b). Recent studies have proven the significance of curcumin at every stage of wound healing. It regulates the inflammatory responses by blocking the production of two essential cytokines, interleukin-1 (IL-1) and tumour necrosis factor alpha (TNF-α) (Kumari et al. 2022). When secreted Wnt proteins attach to a receptor complex, the Wnt/β-catenin signalling cascade is triggered. The pathway's regulation is impacted by the phosphorylation of essential components by casein kinase-1 and GSK3B. Axin and APC bind to b-catenin more tightly as a result of this phosphorylation (MacDonald et al. 2009). The six human isoforms that make up the CK1 family phosphorylate elements of the Wnt pathway and function as both activators and inhibitors. The serine/threonine kinase GSK3B phosphorylates proteins by binding to them, which in turn downregulates β-catenin and may have tumour suppressor properties (McCubrey et al. 2014).

Honey

Honey has long been used as a traditional treatment for wounds and burns, but not much research has been done on the possibility of using it in conventional medicine. Numerous investigations have confirmed the antibacterial properties of honey in vitro, and a limited number of clinical case studies have demonstrated the ability of honey to improve tissue repair and remove infection from highly infected cutaneous wounds (Yaghoobi et al. 2013). Honey's osmotic effects and pH are examples of its physicochemical characteristics, which contribute to its antibacterial activities. Additionally, studies have suggested that honey may have anti-inflammatory properties and boost the body's defences against wounds. Overall, burns, ulcers, and other cutaneous wounds heal more quickly and with less infection. Additionally, it is recognised that specific floral sources in Australia and New Zealand (Leptospermum spp.) produce honeys with stronger antibacterial properties; as a result, these honeys—Medihoney and Active Manuka honey—have been authorised for sale as therapeutic honeys. The information about honey's medicinal qualities is reviewed in this review, which also suggests that honey may be used to treat a wide range of wound types (Yaghoobi et al. 2013). Research conducted in vitro has revealed that honey's immunostimulatory properties on leukocytes result in the synthesis of cytokines, which in turn stimulate and promote cell development. Monocytes were shown to release more tumour necrosis factor alpha (TNF-α) when honey was present at a concentration of 1%0.38 Although it is typically thought to be harmful to stimulate inflammation, honey's anti-inflammatory properties can modulate this inflammatory response (Tonks et al. 2001). Therefore, there was no increase in TNF-α release when 1% honey was administered to inflammatory macrophages that were created by activating monocytes with lipopolysaccharide and opsonised zymosan. Additionally, the honey reduced the formation of reactive oxygen intermediates that were created during the respiratory burst. In a related investigation, it was discovered that 1% of honey stimulated the production of TNF-α, IL-1β, and IL-6 from monocytes—cytokines that are known to be involved in tissue repair in vivo (Tonks et al. 2001).

Gotu Kola, or Centella asiatica

Centella asiatica (CA) is a clonal, perennial herbaceous creeper that grows up to 1800 m in elevation in damp areas of India. It is a member of the Umbellifere (Apiceae) family. This plant, which grows near water, has no flavour or smell. It grows small oval fruit and has small fan-shaped green leaves. Its flowers are white, light purple to pink, or both (Student 2024). The entire plant is utilised in medicine. In India, China, and Indonesia, gotu kola (Centella asiatica) has been utilised for thousands of years to treat a wide range of ailments. It was used to treat skin diseases including psoriasis and leprosy, mend wounds, and enhance mental clarity. Gotu kola is a common ingredient in wound healing lotions. Find out from your healthcare physician if one is appropriate for you. In addition to dried herbs, tinctures, capsules, tablets, and ointments, gotu kola can be found in teas. Products must be used before the labelled expiration date and kept in a cool, dry location. There is a chance that gotu kola could damage the liver (Singh et al. 2010). Gotu kola-containing wound dressings promoted healing in a variety of wound types. This comprises diseased tissue, uneven tearing from blunt-force trauma, and clean wounds from sharp objects (Tcheng et al. 2021). Asiaticoside, madecassoside, and asiatic acid, three of Centella Asiatica's active ingredients, interact with different signalling pathways and receptors to facilitate wound healing. These substances upregulate TGF-β1 (Transforming Growth Factor-beta 1) expression, which is essential for collagen synthesis and fibroblast proliferation. Additionally, they increase the expression of VEGF (Vascular Endothelial Growth Factor), which attaches to endothelial cell receptors to promote angiogenesis. Furthermore, promoting the healing process are Centella Asiatica's anti-inflammatory and antioxidant qualities, which lessen oxidative stress and inflammation. Centella Asiatica does not bind to a single particular receptor, but rather interacts in a variety of ways that are important for efficient wound healing (Diniz et al. 2023; Bandopadhyay et al. 2023).

Wound healing activity

Wound healing is a complex process that involves four sequential phases; inflammation, proliferation, maturation and ends with scar formation. Reactive oxygen species (ROS) play a pivotal role in the orchestration of the normal wound healing response (Moura et al. 2021). In a research work by FBR de Moura et al. (2021), after three and seven days of treatment, wounds treated with MiEH exhibited faster closure (Fig. 6A, B). Following a 3-day course of therapy, compared to the CO group (10.3% ± 1.1), the percentages of wound area closure were higher in wounds treated with 2% MiHE (21.1 ± 1.1) ($p \leq 0.05$), 4% MiHE (20.8% ± 1.8) ($p \leq 0.05$), and 6% MiHE (30.2% ± 2.6) ($p < 0.01$). Following a week of therapy, the wound closure improved with all concentrations of MiHE ($p < 0.05$), with closures of 69.8% ± 2.5 (2% MiHE), 67.9% ± 1.2 (4% MiHE), and 69.5% ± 3.2 (6% MiHE) compared to the 60% ± 2.6 of the CO group. Solely lesions administered with 4% ($p \leq 0.05$) and 6% ($p \leq 0.01$) MiHE were different after 5 days of therapy (40.8% ± 1.7 and 44.5% ± 2.6 of wound closure, respectively) from the wounds of the CO group (31.1% ± 2.6). However, 4% MiHE proved effective in hastening the closure of wounds. 4% MiHE showed pro-fibrogenic, pro-angiogenic, and anti-inflammatory properties (Table 1). Furthermore, as MMP-9 is essential for both the keratinisation process and epidermal closure, an increase in MMP-9 aids in the progression of the proliferative stage. Accordingly, this study has demonstrated that 4% MiHE has advantageous qualities and could be a viable choice for quickening the healing of skin wounds. Copyright Elsevier, Reprint with permission from FBR de Moura et al. (2021).

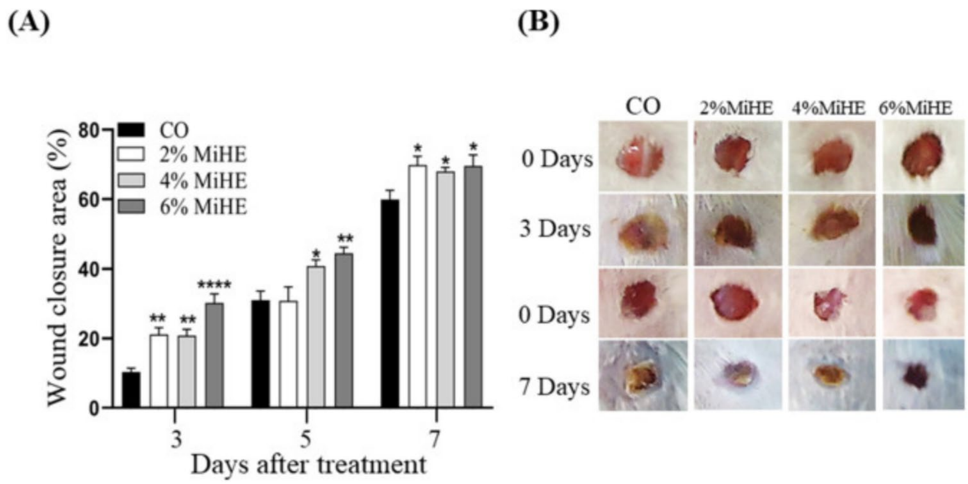


Fig. 6 Region of wound closure. **A** After three and seven days of treatment, the percentage of wound closure in animals injured and treated with vehicle (group CO) and animals treated with M. ilicifolia hydroethanolic extracts at doses of 2% (2% MiHE), 4% (4% MiHE), and 6% (6% MiHE) is graphically represented. **B** Macroscopic pictures of the wound closing on day 0 following wound induction and on days 3 and 7 following treatment. One-Way ANOVA and Bonferroni post-tests (n = 8) were used to statistically evaluate the data, which are displayed as mean and standard error. When $p < 0.05$, $^{**}p \leq 0.01$ and $^{****}p \leq 0.0001$, there is a statistically significant difference with respect to the CO group. . Copyright Elsevier, Reprint with permission from FBR de Moura et al. (2021)

In another study by Bakr et al. (2021) the rate of wound healing in male Wister rats treated once, on the wound induction day, with low to medium concentrations of *H. syriacus* sponge was significantly faster compared to both the control, non-treated group, (SPC-low, $p = 0.026$ and SPC-medium, $p = 0.037$) and the standard* treated group (SPC-low, $p = 0.031$ and SPC-medium, $p = 0.026$). Although the SPC-high didn't show a significant change compared to both the control or standard treated group, it showed a higher healing rate as shown in Fig. 7 (Bakr et al. 2021).

Comparative study of plant derived natural medicine and animal derived natural medicines

As aspect of Pharmacokinetics study of Plant-Derived Natural Medicines & Animal-Derived Natural Medicines; Plant-derived natural medicines have strong therapeutic effects but suffer from bioavailability challenges, but natural medicines. Animal-derived have superior bioavailability and effectiveness, particularly in advanced wound healing, due to their structural compatibility with human tissue (Table 2). There have many reasons in Plant-Derived Natural Medicines have low Bioavailability due to enzymatic degradation, poor solubility and rapid clearance but Animal-Derived Natural Medicines have higher bioavailability, peptides and structural proteins mimic natural ECM components (Jain et al. 2021b; Rathore et al. 2014).

Clinical trials

Table 3 summarizes some clinical trials contrasting different wound healing therapy methods in different age ranges and administration routes. Antimicrobial topically applied silver sulfadiazine was employed in a Phase III trial in adults, and efficacy was demonstrated for infection prevention and wound healing. Platelet-derived Growth Factor (PDGF), a second agent that is applied topically, weekly in a Phase II study, showed extraordinary healing of wounds and granulation tissue formation. In children, the application every 2–3 days with hydrogel dressings was observed to enhance healing. In surgical wound infections, intravenous administration of vancomycin every 12 h in Phase II showed reduced incidence of infection. Oral administration of omega-3 fatty acids 1–2 g daily was tested in a Phase III trial for its application in inflammation reduction and healing of chronic wounds. Finally, stem cell therapy, tested in aged subjects in a Phase I trial, was also found to heal chronic and non-healing wounds. These research studies depict varied practices in wound care, from traditional antimicrobials to new regenerative therapies.

Conclusion

The pharmacological value of plant drugs as healing agents of wounds has emerged increasingly into prominence due to their ability to alter molecular pathways and receptor targets that are involved in tissue repair. The review herein identifies the relevance of important

Table 1 Herbal drugs with their mechanism and receptor targeting in the treatment of asthma

Herbal drug	Family	Botanical name	Part of plant	Mechanism of action	Receptor	References
Thyme	Lamiaceae (Mint family)	*Thymus vulagris*	Leaves and stems	Antimicrobial,antifungal, antifungal and antiviral properties due to the presence of phenolic compounds like thymol and carvacrol	Thymol and carvacrol intract with the bacterial cell membrane, TLR4, TGF-Beta,PDGF,FGF,VEGF	Vassiliou et al. 2023, Nadi et al. 2023)
Guggul	Burseraceae	*Commiphora wightii*	Resin(gum) from the bark	Antioxidant:scavenges free radicals,Anti-inflammatory,hypolipidemic,Anti-atherosclerotic	Proliferator-Activated Receptor gamma, Toll-like receptor 4, liver X receptor-α	Kunnumakkara et al. 2018)
Haridra	Zingiberaceae	*Curcuma longa*	Rhizome (root)	Anti-inflammatory,NF-kB activation,Antioxidant,Anticancer, Anti-diabetic	NF-Kb (nuclear factor kappa B), PPAR-γ(Peroxisome proliferator-Activated receptor gamma),sirtuin 1,VEGFR,TRPV1	Mazidi et al. 2016)
sesame	Pedaliaceae	*sesamum*	seeds	Anti-inflammatory inhibits prostaglandin synthesis, Antioxidant, Anti-bacterial	Peroxisome proliferator-activated receptor gamma(PPAR-γ), Toll-like receptor(TLR4),Nuclear factor kappa B(NF-Kb)	Hadipour et al. 2023)
Comfrey	Boraginaceae	*Symphytum officinale*	Root, Leaf	Anti-inflammatory, promotes cell proliferation and wound contraction	Toll-like receptor(TLR4),Fibroblast growth factor(FGF),Transforming growth factor -beta	Mahmoudzadeh et al. 2022)
Chamomile	Asteraceae	*Matricaria chamomilla*	Flower	Anti-inflammatory, antimicrobial, promotes granulation tissue formation	VEGF receptor, Cannabinoid receptor (CB1And CB2),Adenosine receptor, GABA receptor	Singh et al. 2011)
Tea Tree Oil	Myrtaceae	*Melaleuca alternifolia*	Leaves	Antimicrobial, anti-inflammatory, promotes wound debridement	Toll-like receptor,4, Transient receptor potential vanilloid 1,VEGF,mAChR, GABA receptor	Carson et al. 2006)
Arnica	Asteraceae	*Arnica montana*	Flower	Anti-inflammatory, antimicrobial, reduces swelling		Smith et al. 2021)
Neem	Meliaceae	*Azadirachta indica*	Leaves, Bark	Antimicrobial, anti-inflammatory, promotes wound healing	TLR4,TGF-Beta, (PDGF), Fibroblast growth factor(FGF),Vascular endothelial growth factor(VEGF),Nuclear factor kappa B(NF-KB)	Alzohairy 2016)
Echinacea	Asteraceae	*Echinacea purpurea*	Root, Aerial parts	Antimicrobial,anti-inflammatory, promotes tissue regeneration	TLR4, TGF-Beta receptor, PDGFR, FGFR, VEGFR	Burlou-Nagy et al. 2022)

Table 1 (continued)

Herbal drug	Family	Botanical name	Part of plant	Mechanism of action	Receptor	References
Plantain	Plantaginaceae	*Plantago major*	Leaves	Anti-inflammatory, antimicrobial, promotes tissue regeneration	TLR4, TGF-βR,PDGFR, VEGFR,CB1,CB2 RECEPTOR	Zhakipbekov et al. 2023)
Eucalyptu	Myrtaceae (Myryle family)	*Eucalyptus globulus*	Leaves	Anti-inflammatory,antimicrobial,and antioxidant properties due to the presence of compounds like eucalyptol (1,8-cineole) and flavonoids	Eucalyptol interacts with the transiter receptor,(TRPA1) Receptor	Farman 2021)
Garlic	Amaryllidaceae	*Allium sativum*	Bulb	Antimicrobial, anti-inflammatory, enhances wound healing	Nitric oxide synthesis,(TLRs),toll like receptor,transforming growth factor -beta	Batiha, et al. 2020)
Lavender	Lamiaceae	*Lavandula angustifolia*	Flowers	Antimicrobial, anti-inflammatory, promotes tissue regeneration	Adenosine receptors,TRPV1 Receptor,Estrogen receptor	Miastkowska et al. 2021)
Rosehip Oil	Rosaceae	*Rosa canina*	Seeds	Promotes collagen production, anti-inflammatory, antioxidant	(TRPV1) Transient receptor potential vanilloid 1,Gamma-Aminobutyric acid	Mármol et al. 2017)
SeaBuckthorn	Elaeagnaceae	*Hippophae rhamnoides*	Berries	Anti-inflammatory, antimicrobial, promotes tissue regeneration	FGF-2(Fibroblast Growth Factor-2)receptor,TRPV1 Receptor	Suryakumar and Gupta 2011)
St. John's Wort	Hypericaceae	*Hypericum perforatum*	Flowering tops	Anti-inflammatory, antimicrobial, promotes granulation tissue formation	5-HT1A AND 5-HT2A Receptor, Vascular Endothelial Growth Factor Receptor,TRPV1	Wölfle et al. 2013)
Witch Hazel	Hamamelidaceae	*Hamamelis virginiana*	Bark, Leaves	Anti-inflammatory, astringent, antimicrobial	NF-Kb,STAT6,TNFR,IL-4R,IL-6,TSLP Receptor	
Marshmallow Root	Malvaceae	*Althaea officinalis*	Root	Anti-inflammatory, promotes tissue regeneration	Toll-like receptor 4, IL-1R(Interleukin-1 receptor),TNFR(Tumor necrosis factor receptor),FGF-2,FEGF(Vascular endothelial growth factor)	Khmiyas et al. 2015)

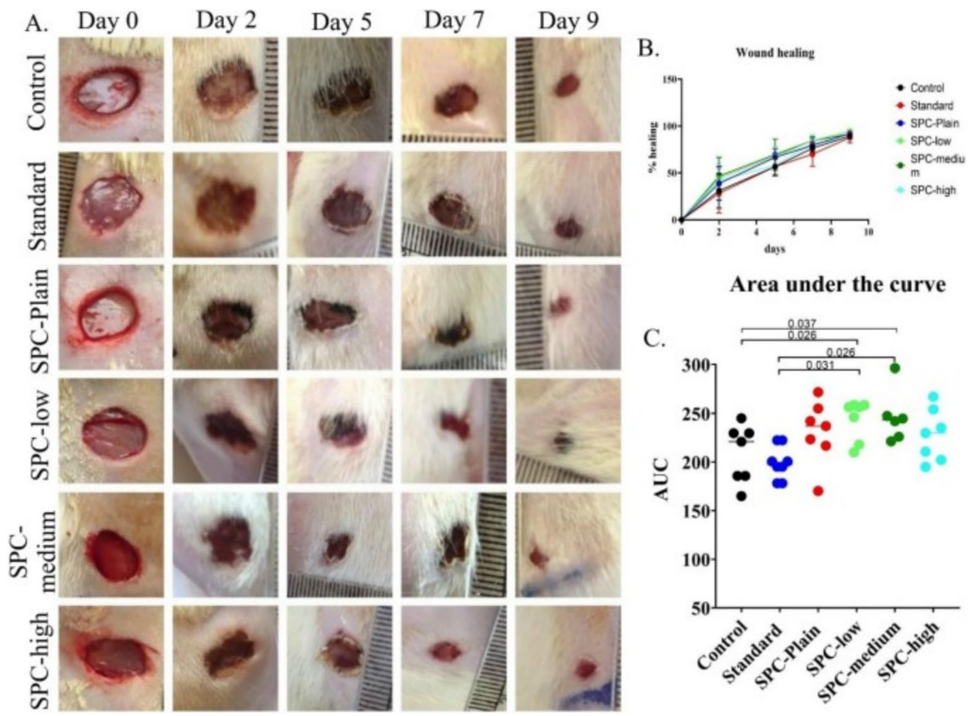


Fig. 7 Different formulae's healing effects in comparison to a commercially available standard medication (Mebo®). The rate of wound healing from day 0 (the day of wound induction) to day 9 is displayed in the left panel (**A**). Area under the curve (AUC) is quantitatively measured in the right panel (**B**). Each individual wound's data are displayed as a scattered plot, and the mean ± SD of at least 7 wounds is displayed. $P < 0.05$ for SPC-low and medium as compared to control and standard; the precise P value is displayed in the graph. Copyright Elsevier, Reprint with permission from Bakr et al. (2021)

Table 2 Comparative studies on plant and animal derived medicines in treatment wound

Aspect	Plant derived natural medicines	Animal derived natural medicines	References
Anti-inflammatory	Curcumin (turmeric), aloe vera, calendula reduce pro-inflammatory cytokines (TNF-α, IL-6)	Honey, fish skin collagen, and silk fibroin modulate inflammation and enhance macrophage activity	Peng et al. 2021)
Antioxidant activity	Polyphenols and flavonoids (e.g., in green tea, turmeric) scavenge free radicals	Peptides and enzymes (e.g., lactoferrin from milk, lysozyme in honey) reduce oxidative stress	Frei and Higdon 2003)
Antimicrobial action	Essential oils (e.g., tea tree oil), tannins (e.g., pomegranate) disrupt bacterial membranes	Honey (hydrogen peroxide production), chitosan (from crustacean shells) exhibits bacteriostatic properties	Swamy et al. 2016)
Collagen synthesis & Tissue regeneration	Aloe vera, Centella asiatica promote fibroblast proliferation	Fish skin collagen, silk fibroin directly contributes to extracellular matrix (ECM) remodeling	Lu et al. 2004)
Moisture retention & Barrier protection	Mucilage (e.g., aloe vera, plantain) forms a protective film	Snail mucin, gelatin, and fish collagen provide hydration and protective scaffolding	Thomas et al. 2020; Liang et al. 2021)

receptors for growth factors and neurotransmitter receptors in controlling the different processes involved in the wound healing response. Medicinal plants have also been shown to exhibit excellent wound-healing capacity with their antioxidant, anti-inflammatory, and antimicrobial activities. Furthermore, comparison of plant-derived natural medicines versus animal-derived natural medicines indicates that though plant-derived drugs are subjected to bioavailability concerns, they pose promising non-surgical therapy regimens for non-healing and chronic wounds. Molecular insights and clinical studies of herbal medicine may provide the basis for creating new, receptor-targeted therapeutics that will eventually enhance wound care and management in clinics.

Table 3 Clinical trial studies on wound healing treatments

S. no	Ages	Phase	Route of administration	Drug	Dose/frequency	Findings	Clinical Trials. Gov ID
1	Adults	III	Topical	Silver sulfadiazine	Apply 1–2 times daily	Effective in preventing infection; promotes healing	NCT01390519
2	Adults	II	Topical	Platelet-derived growth factor (PDGF)	Administer weekly	Significant improvement in wound closure and granulation tissue	NCT00428795
3	Children	I	Topical	Hydrogel dressings	Change every 2–3 days	Improved healing in pediatric wounds	NCT03463514
4	Adults	II	Intravenous	Antibiotic (e.g., vancomycin)	Every 12 h	Reduced infection rates in surgical wounds	NCT02475103
5	Adults	III	Oral	Omega-3 fatty acids	1–2 g daily	Enhanced healing and reduced inflammation in chronic wounds	NCT02197821
6	Older Adults	I	Topical	Stem cell therapy	As per protocol	Promising results in chronic and non-healing wounds	NCT02771157

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Declarations

Conflict of interest We declared no conflict of interest.

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